

The Local Compressor

Micro View - Causal Base

Input Video Segment



Vision Encoder (e.g., SigLIP)

Visual Tokens ($\mathbf{X}_i$)

Small Language Model (SLM)

Causal Attention: M attends to all conditions

System Prompt

Visual Tokens $\mathbf{X}_i$

User Query Q

Memory Tokens M ($k_{max}=128$)

Full Memory Tokens (k_{max})



Inference-Only Bypass

Training-Free ATA

$\mathbf{h}_{rel}$

Zero-shot Relevance

$$s_i = \sigma \left((\mathbf{w}_{yes} - \mathbf{w}_{no})^\top \mathbf{h}_i^{rel} \right) \in (0, 1)$$

ATA Controller

Global Budget B_{max}

Segment 1: High Score s_i



Keep all details ($k_i = 128$)

Segment 2: Low Score s_i



Keep temporal anchors ($k_i = 4$)

The Global Decoder

Macro View - Time-Aware Sequence

$\langle t=0.0s \rangle$



$\langle t=2.0s \rangle$

$\langle t=4.0s \rangle$

User Query Q

Large Language Model (e.g., Qwen3-4B)

Global Self-Attention

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Global Self-Attention

Final Answer